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# Two Regimes of the Chiral Mass Coupling

## Quantum Measurement as Bath-Induced Synchronization


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## Abstract

We read quantum measurement as a synchronization process and place the reading squarely in the single-world, $\psi$-ontic, nonlocal family of interpretations — alongside Bohmian [2, 3] and Nelsonian [19] mechanics, and in the lineage of relational-clock approaches to quantum time [29, 36]. The mechanism is grounded in the Dirac equation: in the chiral (Weyl) basis the mass term is the off-diagonal coupling between left- and right-handed sectors. A Madelung reduction (§2.2) shows that in a *closed* system this coupling generates unitary normal-mode (Rabi) precession with **no attractor** — it cannot phase-lock. This is a consistency (no-go) result, not a derivation of superposition's survival: a coupling that superficially resembles a Kuramoto synchronizer is shown to have an off-switch, and that switch is the closed/open boundary. An attractor appears only when the system is *opened* to a dissipative bulk and the bath is traced out, where the reduced phase obeys an Adler/Kuramoto equation [9]. Measurement is that opening: a fast resonant capture (Stage 1) followed by irreversible equilibration to the bulk's collective reference (Stage 2). The reversible→irreversible boundary is exhibited directly by the nuclear-spin echo, the catch-and-reverse of a superconducting qubit [14], and Stern–Gerlach recombination. We model the bulk reference honestly as a *faint* statistical bias (parts per million, as in MRI) on an overwhelmingly thermal reservoir, not a coherent macrostate.

We are explicit about scope. The framework does not modify quantum mechanics: it accepts Bell's theorem, takes the correlations from the entangled state (a clock-as-local-variable model is sub-classical, CHSH $\leq \sqrt{2}$; Appendix A), and preserves no-signaling — confining its contribution to the local measurement mechanism. It carries a physical preferred frame that enters only through Stage 2. Two problems separate it from a complete theory and are stated as such: a derivation of the Born measure, and an explicit dynamical law for the background-field selection — which is also the locus of the framework's nonlocality. Its one candidate signature, a linewidth-dependent gravitational Bell effect (§6), does *not* follow from the framework plus standard QED but requires an additional, non-covariant postulate (§6.2, Appendix C); that experiment therefore tests the postulate, not the framework's core. We present MCI as a physically-motivated single-world *interpretation* with a concrete measurement mechanism — not as a theory making novel predictions.

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# 1. Introduction

### 1.1 The measurement problem

The Schrödinger equation is linear and deterministic — as is its relativistic counterpart, the Dirac equation on which we build — yet measurements have single, definite outcomes. A system in superposition $\psi = \alpha|0\rangle + \beta|1\rangle$ entangled with a detector evolves unitarily into

$$|\Psi\rangle = \alpha|0\rangle|D_0\rangle + \beta|1\rangle|D_1\rangle,$$

and the equation contains no mechanism that selects one branch. The Born rule $\mathrm{P}(0) = |\alpha|^2$ is postulated, not derived. Copenhagen asserts a collapse with no mechanism; Everett [1] keeps unitarity at the cost of unobservable branches; Bohmian mechanics [2, 3] accepts explicit nonlocality through a pilot wave.

### 1.2 The gap, and the proposal

Environmental decoherence [4, 5] successfully describes *what happens* when a superposition meets a macroscopic environment — the suppression of off-diagonal density-matrix elements and the emergence of pointer states. What it does not supply is a *dynamical mechanism*: the environment is treated as an abstract bath of many modes, and decoherence presupposes the Born trace rule to compute the reduced state rather than producing it.

This paper proposes a mechanism. The relevant phase degrees of freedom are not added by hand; they are the chiral phase sectors already present in the Dirac equation (§2). The central observation is that the *same* off-diagonal coupling behaves in two sharply different ways depending on boundary conditions: closed and unitary, it precesses as coherent normal modes and protects superposition (§2.2); opened to a dissipative bulk, it acquires an Adler/Kuramoto attractor and locks (§2.3). Measurement is the transition from the first regime to the second — a resonant capture followed by irreversible equilibration to a macroscopic reference (§3). We call the resulting reading the **Many Clocks Interpretation (MCI)**: every particle carries an internal phase clock, dissipative interactions synchronize clocks locally, no observer is required, and no branching occurs.

### 1.3 Scope and honesty (stated once, up front)

To keep the argument disciplined we state the boundaries here rather than distributing them through the paper:

- **We do not modify quantum mechanics.** The Dirac equation, the Born rule's numerical content, and Bell's theorem are unchanged.
- **The correlations are standard quantum mechanics, not the framework.** Bell correlations [6] come from the entangled state and the full spinor (or polarization) Hilbert space. We show in Appendix A that a clock-as-hidden- variable model of the *outcomes* gives CHSH $\leq \sqrt{2}$ — sub-classical — so the correlation *must* come from the standard structure, and the framework's contribution is confined to the local measurement mechanism.
- **We accept Bell and preserve no-signaling.** The framework is not superdeterministic; detector settings are free, and nothing in it permits faster-than-light signaling (§7.5).
- **The Born rule is reframed, not derived.** §7.3 and §8 isolate the two gaps honestly.
- **One candidate prediction, contingent on a postulate.** A linewidth-dependent gravitational Bell effect (§6). We show it does *not* follow from the framework plus standard QED (§6.2, Appendix C); it requires an additional, non-covariant coupling. The experiment is a clean falsification test of that postulate, and is the framework's one observational handle.
- **The framework carries a preferred frame, and we say so.** In a single world, Bell nonlocality is anchored to a physical preferred frame — the bulk/vacuum rest frame — which enters only through measurement (Stage 2); the unitary dynamics stays exactly Lorentz covariant. Ordinary matter couples to that frame, beyond the electromagnetic locking, only gravitationally and so weakly. Demonstrating quantitatively that this evades the precision Lorentz tests is part of what the framework still owes (§8).
- **The chiral coupling K = m is a fixed constant.** It is set once by the Higgs vacuum expectation value, a Lorentz scalar; gravity *reads* the chiral register (the mass term $\bar{\psi}\psi$ to which the stress tensor couples) but does not *modulate* K. There is no K(Φ) dependence, so the framework carries no equivalence-principle liability — only the preferred-frame (Lorentz) liability above.

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## 2. Two regimes of the chiral mass coupling

### 2.1 Chiral decomposition

In the Weyl basis the Dirac spinor splits as $\psi$ = ($\psi$_L, $\psi$_R)$^{\mathrm{T}}$ and the Dirac equation [7] becomes the coupled pair

$$i\bar{\sigma}^{\mu}\partial_{\mu}\psi_L = m\,\psi_R, \qquad i\sigma^{\mu}\partial_{\mu}\psi_R = m\,\psi_L,$$

with $\sigma^{\mu} = (I, \vec{\sigma})$, $\bar{\sigma}^{\mu} = (I, -\vec{\sigma})$. For m = 0 the two equations decouple and the particle has definite chirality; for m $\neq$ 0 the mass couples the sectors. In the chiral basis the mass is the single off-diagonal entry of the Dirac Hamiltonian,

$$H = \begin{pmatrix} \vec{\sigma}\cdot\vec{k} & m \\ m & -\vec{\sigma}\cdot\vec{k} \end{pmatrix},$$

set by the Higgs–Yukawa coupling, m = y_f$\langle\varphi\rangle$ (in SI units the Compton frequency $mc^2/\hbar$). This identification is textbook; what follows is a reading of its dynamical consequences, and the central point is that the mass coupling is *not* itself a synchronization attractor.

### 2.2 The closed system: unitary precession, no attractor

Apply the polar (Madelung) decomposition $\psi$_X = $\sqrt{\rho_X}$ · e^{i$\varphi$_X} · $\chi$_X (X = L, R), with $\rho$_X ≥ 0 a real amplitude density, $\varphi$_X a real phase, and $\chi$_X a unit two-spinor carrying the spin orientation. In the rest frame ($\nabla \to 0$, $\chi$ constant), separating real and imaginary parts of the coupled equations gives

$$\dot{\phi}_L = -m\sqrt{\rho_R/\rho_L}\,\cos\Delta, \qquad \dot{\rho}_L = +2m\sqrt{\rho_L\rho_R}\,\sin\Delta, \qquad \Delta \equiv \phi_R - \phi_L,$$

and the mirror equations for R. Two features are decisive. First, the trigonometric functions land in the *opposite* places from the Kuramoto model [8]: the **sine sits in the amplitude equation and the cosine in the phase equation**, whereas Kuramoto carries its sine in the phase equation. Second, the phase-difference dynamics

$$\dot{\Delta} = -m\cos\Delta\,\frac{\rho_L - \rho_R}{\sqrt{\rho_L\rho_R}}$$

vanish on the symmetric manifold $\rho$_L = $\rho$_R for *every* $\Delta$. There is no restoring force and no preferred phase difference — a marginal line of fixed points, not a basin. The motion is normal-mode (Rabi) precession: defining $\psi$_± = ($\psi$_L ± $\psi$_R)/$\sqrt{2}$ gives i$\partial t$ $\psi$± = ±m $\psi$_±, so the chiral populations flop coherently at beat frequency 2m.

It is worth making explicit *why* the absence of a basin protects superposition, because the two statements are one fact in different language. Everything physical about a chiral superposition lives in the relative phase $\Delta$ (and the amplitude ratio); to *lock* is to have a restoring force that pins $\Delta$ to a preferred value $\Delta$*, and to pin $\Delta$ is precisely to destroy the superposition. The contrast with Adler/Kuramoto is exact: there $\dot{\Delta}$ = $\Delta\omega$ − K sin $\Delta$ derives from a potential V = −K cos $\Delta$ with a genuine *minimum* (a basin, a restoring force), whereas the cosine-in-phase / sine-in-amplitude pairing above is the signature of a *conservative* (Hamiltonian) flow, not a *gradient* (relaxational) one — it conserves a quantity rather than descending a potential. Geometrically, the mass term (a $\sigma$_x coupling) precesses the Bloch vector in closed circles *on the surface* of the sphere at the rate 2m; locking would require it to spiral inward (toward a mixed state) or into a pole (a definite outcome), and unitary precession does neither, conserving purity exactly. (When the system is opened, that inward spiral acquires an explicit rate — $\Gamma/2 = -\mathrm{Im}\,\Sigma$, the imaginary part of the dressed Dirac mass — derived in §3.6.)

This is not a missing feature; in the closed regime it *is* the physics. A closed, unitary system admits no Lyapunov function for an order parameter to climb — Liouville's theorem makes the flow volume-preserving, so it has no attractors of any kind. The chiral phases can only precess, never settle. No relabeling rescues an attractor: redefining $\psi$_R → i$\psi$_R moves the sine into the phase equation, but its prefactor [$\sqrt{\rho_L/\rho_R}$ − $\sqrt{\rho_R/\rho_L}$] is odd under L ↔ R and vanishes on $\rho$_L = $\rho$_R, so no single-signed K sin $\Delta$ term survives. An attractor cannot arise from unitary evolution; a bath must be traced out.

We are careful about what this establishes. That an *isolated* system preserves superposition is, on its own, just unitarity — true of any closed system, and not something the chiral reduction is needed to prove; reading §2.2 as a *derivation* of superposition's survival would be claiming credit for Liouville's theorem. The content here is narrower and framework-specific. The mass term couples two phase sectors and superficially *resembles* a Kuramoto synchronizer, so one might fear it carries a built-in, always-on lock — in which case every massive particle would continuously self-measure even in perfect isolation, contradicting observed coherence and sinking the framework. The reduction shows it does not: the would-be synchronizer has an off-switch, and that switch is the closed/open boundary. This is therefore a *consistency (no-go) result* — the proposed collapse mechanism provably does **not** misfire in the one regime where collapse must not occur — rather than an explanation of why superposition survives.

Two scopes must stay distinct. The result above concerns the *chiral $L/R$ relative phase of a single particle*. The *measurement* superposition (system $\otimes$ pointer) is protected by the same general principle — closed $\Longrightarrow$ no attractor — but that protection follows from unitarity at the composite level, not from the chiral calculation specifically; we do not claim the single-particle reduction explains pointer coherence. Finally, the marginal line is not a dead end but the seam to §2.3: neutral stability means a small perturbation moves $\Delta$ freely along that line, neither resisted nor amplified — exactly the soft direction a bath couples to. Opening the system tilts the flat marginal line into a sloped potential with a minimum, the Adler basin of §2.3, so the off-switch and the on-switch are the same structure under two boundary conditions.

### 2.3 The open system: dissipation supplies the attractor

Couple the chiral pair (or a composite particle) to a bulk and trace the bulk out. The reduced phase obeys an Adler equation [9],

$$\dot{\phi} = \omega + K_{\text{eff}} \sin(\Phi_{\text{bulk}} - \phi),$$

which *does* possess a genuine attractor, because the bath now carries away the phase difference and a Lyapunov function exists.

The form above is the lock's **first-harmonic limit**, with the bulk entering as a single biased reference phase $\Phi_{\text{bulk}}$; it is monostable — one locked phase, one outcome. The general dissipative coupling carries a second term. A bulk that *measures* the chiral interference channel — a symmetric, $\langle\cdot\rangle$-free coupling even under $\phi \to \phi + \pi$ — contributes a second harmonic, so

$$\dot{\phi} = \omega - \gamma \sin 2\phi - \epsilon \sin(\phi - \Phi_{\text{bulk}}),$$

with $\gamma$ the symmetric measurement rate and $\epsilon$ the polar bias of the line above. The anti-phase fixed point $\phi = \pi$ is stable iff $\epsilon < 2\gamma$: for a *faint* bias ($\epsilon \ll \gamma$ — the ppm thermal bias of §4.1) the lock is **bistable**, stabilizing phases near 0 and $\pi$ (a binary), the bias merely deepening one well; for a *dominant* bias ($\epsilon > 2\gamma$) it collapses to the monostable first-harmonic lock. Both are limits of one equation. The two-outcome structure of a measurement is thus the symmetric (measurement) part of the bulk coupling; the bias selects among the basins but does not create them. This *stabilizes* a binary but fixes neither its cardinality (set by the measured observable's spectrum) nor the Born weights (§8), both of which remain open.

This — not the closed mass term — is the framework's synchronization step, and it places the picture within the established literature on quantum synchronization [10–13], where Kuramoto-form phase dynamics are derived from the semiclassical limit of quantum master equations for genuinely self-sustained (limit-cycle, dissipative) oscillators. The two regimes are the same off-diagonal coupling under two boundary conditions: closed $\to$ precession (superposition preserved); open $\to$ Adler lock (measurement). Coupling and synchronization are therefore not rival mechanisms but the same coupling under two boundary conditions. The off-diagonal coupling is continuous across both regimes; the *attractor* is not — it is supplied by the bath. Where there is no bath the coupling can only precess; once a bath is present and traced out, the dissipation converts that precession into a locking attractor. The boundary between them — the onset of dissipative coupling to a bulk of overwhelming inertia — is the framework's candidate for a physical Heisenberg cut, made precise as a condition on the effective action in §3.5. We use "synchronization" hereafter only in this strict dissipative sense.

### 2.4 Normal modes and the Zitterbewegung beat

Reinstating the kinetic term, diagonalizing H on a helicity block gives eigenvalues $\pm\sqrt{k^2+m^2} = \pm E$: the relativistic dispersion. The normal-mode splitting is 2E, equal to 2m at rest, and this is the Zitterbewegung beat.

We are explicit about a factor of two that is easily blurred. The Zitterbewegung beat is the splitting between the **positive- and negative-energy normal modes** (+E and −E), which at rest is $2mc^2/\hbar$. It is *not* the difference of a "temporal clock" at $\omega$_t = E/$\hbar$ and a "spatial clock" at $\omega$_s = p·v/$\hbar$; in the rest frame that difference tends to $mc^2/\hbar$, not $2mc^2/\hbar$, so the temporal/spatial pairing does not reproduce the beat. The two clocks whose interference gives Zitterbewegung are the ±E modes; the de Broglie carrier is their symmetric mode at $\omega$ = E/$\hbar$. The Zitterbewegung beat is the empirical signature that the chiral coupling is real: without m, $\psi$_L and $\psi$_R free-stream and no beat forms.

| Claim | Status |
|---|---|
| Mass couples $\psi$_L ↔ $\psi$_R; m = 0 decouples them | Standard QFT; textbook |
| Higgs–Yukawa sets the coupling, m = y_f⟨$\varphi$⟩ | Standard Model |
| Closed mass coupling = unitary precession, **no attractor** | This work (rigorous; §2.2) |
| Measurement = dissipative Adler/Kuramoto lock to a bulk | This work (interpretive; §3) |

### 2.5 When the open regime engages: free flight, boundaries, and gradients

The two regimes raise a question §2.2–2.3 leave open: at any given moment, which one is acting? The answer is set not by time but by *place*. The open, dissipative regime — the only one we call synchronization — engages only where an external structure **deforms the wave's amplitude**: a material boundary (an aperture edge, a detector surface), a field gradient (Stern–Gerlach, MRI), or spacetime curvature (a tidal gradient). In free flight through an unbiased, gradient-free vacuum, only the closed regime runs, and the particle stays coherent.

This is not a postulate but a consequence of what decoherence requires. A freely propagating *inertial* particle does not radiate; it deposits no real quantum in the vacuum, so there is no record to carry away which-path information and nothing to decohere it — which is why interferometers hold coherence over macroscopic paths. The one vacuum interaction always present in flight, the self-energy dressing that renormalizes the mass, is coherent and non-recording: it *maintains* $K = m$ rather than reading it. A record — and therefore a measurement — forms only where the particle accelerates or scatters, that is, at a boundary.

In flight, then, the dynamics is the closed-regime guidance already contained in the Madelung decomposition of §2.2, now read ontologically: the phase guides through $\mathbf{v} = \nabla\varphi/m$, with the amplitude entering through the quantum-potential term (§7.3). We adopt this guidance law — it is the dynamical guiding law §7.3 records as owed — but locate the *outcome-selecting* randomness not in flight but in the unbiased background sampled **at the boundary** (§3.3); the $|\psi|^2$ form of that selection remains open (§8), with relaxation to quantum equilibrium [42, 43] the candidate route. One caution against a hidden assumption: the in-flight guidance phase and the internal Zitterbewegung clock are not independent factors but the two normal modes of the $K = m$ coupling (§2.4) — the de Broglie carrier and the $\pm E$ beat. Treating their rates as decoupled is a rest-frame approximation, not a separate premise.

Boundaries come in two kinds, distinguished by whether a record is taken. A **coherent** boundary — an aperture, or a Stern–Gerlach gradient — reshapes or splits the wave while taking no which-path record; it stays in the closed regime, which is exactly why two-slit interference survives the slits and a Stern–Gerlach beam can be recombined. A **recording** boundary — a detector surface, an absorbing screen, a readout coil — exchanges real quanta and fires the open regime. A field gradient (Stern–Gerlach, MRI) is a *biased* boundary: an applied field, not the unbiased vacuum, sets the reference axis, yet the irreversible record still forms only at the screen or coil. Gravity sits at the far weak end: a uniform potential is removable by the equivalence principle and does nothing in flight, and only a curvature/tidal gradient is a genuine in-flight gravitational deformation — equivalence-principle-safe, since it does not modulate the fixed $K$ (§1.3), but negligible in the laboratory and relevant only in strong-field astrophysics.

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## 3. Measurement as resonant capture and bulk re-synchronization

### 3.1 Two stages

A measurement separates cleanly into two steps, consistent with the locality of fundamental interactions:

- **Stage 1 — resonant capture.** The incoming system phase-couples to a single bulk-bound partner at the interaction vertex (a photon to a bound electron, a spin to a detector dipole). Crucially, the partner is *never a closed system* — it is coupled to its host atom and the surrounding material — so Stage 1 carries a weak dissipative component from the outset. This is the regime of injection locking and of the laser: constructive (resonant) interaction selects which mode the system locks to, and the weak dissipation lets a locked state begin to form. The capture is governed by the Arnold-tongue condition $|\Delta\omega| \lesssim$ K_pair: mismatches large compared to the coupling are not pulled in.
- **Stage 2 — bulk equilibration.** The perturbed partner re-locks to the bulk's collective reference by *dissipating* its phase mismatch into the local environment. The dissipation, not any amplifier, is what makes this step a lock; it is present in every interaction with an open environment, with or without a detector. The physical channel, however, differs by system. In an ordinary (non-recording) interaction it is **relaxation toward local equilibrium** — fluorescence, phonon emission, the bound electron settling back toward the bulk average exactly as a spin relaxes toward $B_0$ in NMR (T1/T2). In a detector built to *record*, it is **amplification** of the single excitation into a macroscopic cascade (avalanche, PMT gain). Both are dissipative and one-way, but they are not the same physics: amplification only renders the already-dissipated event a permanent, readable record. Irreversibility in either case is the mismatch escaping into a continuum of environmental modes from which it does not recur — relaxation into a genuine bath achieves this, amplification guarantees it. That non-recurrence is irreversibility read as *erasure*: the phase mismatch carrying the system's prior coherence is scattered into the bulk's uncontrolled modes beyond practical recall, so the system cannot recohere and the locked state is stable. The erasure is *effective*, not fundamental — consistent with the no-collapse commitment the global evolution remains unitary and the history is scrambled into the bulk rather than destroyed; what is gone is the system's *locally recoverable* history, the irreversible T1/T2 side of §3.2 and not the refocusable T2* side. Three things are therefore distinct and should not be run together: this *erasure* (irreversibility, which makes the record stable), the *decoherence* it accompanies (the dissipative step yields a *decohered* (mixed) reduced state), and the *selection* of a single definite outcome from that state — the last being the separate role of the background configuration (§3.3), not of the dissipation itself. That an *irreversible act of amplification* is what closes a measurement is an old observation — it is Bohr's irreversibility requirement [21] and the Bohr–Wheeler dictum that "no elementary phenomenon is a phenomenon until it is a registered phenomenon, brought to a close by an irreversible act of amplification" [22] — and the macroscopic-equilibration mechanism we invoke for Stage 2 is closely modeled on the ergodic apparatus-relaxation account of Daneri, Loinger, and Prosperi [23]. Our contribution is not the irreversibility, which is standard, but its identification with a dissipative phase-locking (Adler) step and its separation from the outcome-*selection* role of §3.3.

Both stages are dissipative, separated by a *hierarchy of timescales* — a fast resonant capture and a slower bulk equilibration — rather than by a sharp unitary/dissipative ontological boundary. One might be tempted to treat Stage 1 as strictly unitary and confine all dissipation to Stage 2, but the capture partner is never truly closed; the honest statement is a separation of rates, not of kinds.

### 3.2 The reversible→irreversible boundary is demonstrable

The content of "fast capture, slow equilibration" is that there is a window in which the early dynamics are reversible and a point beyond which they are not. This is not a metaphor; three independent systems exhibit it directly:

- **The nuclear-spin echo.** In NMR/MRI the transverse dephasing that follows a radiofrequency pulse separates into a *reversible* part (T2*, from static field inhomogeneity, refocused by a 180° echo) and an* irreversible* part (T2/T1, stochastic spin–spin and spin–lattice relaxation). The spin echo is a routine, unambiguous reversal of the capture-stage dephasing — a Humpty-Dumpty experiment for spins.
- **Catch-and-reverse of a qubit.** Minev et al. [14] track a quantum jump in a superconducting transmon and reverse it mid-flight with ~82% fidelity, in a regime where the dissipative readout channel has been engineered to near-zero. In framework terms: with the Stage-2 coupling suppressed, no attractor forms and evolution stays reversible until the first real dissipative event.
- **Stern–Gerlach recombination.** Carefully recombined SG paths recover the original superposition: the gradient creates the correlation, not the outcome; the outcome arrives only at the irreversible detection.

### 3.3 Why outcomes are definite — and what this does and does not explain

The detector's macroscopic inertia provides the asymmetry. When a small oscillator (mass m) couples dissipatively to a large one (M $\gg$ m), the small one conforms; the reverse is suppressed by m/M. A single experimental run carries one actual configuration of the background field (§4), and the dissipative flow carries that configuration into exactly one attracting basin. There is then one $\psi$, one background, one outcome, one world — no branching, no projection postulate, no observer.

We are deliberate about what this does *not* settle. The attractor explains why there is exactly **one** outcome per run; it does not explain why the long-run **frequency** of each basin is $|\alpha|^2$. That is the Born measure, isolated as an open problem in §7.3 and §8. Conflating the two — treating "definite outcome" and "correct weight" as one result — is the error we most want to avoid.

### 3.4 Decoherence as synchronization to the bulk

In the standard account a measured system loses *quantum* coherence with its entangled partner — the off-diagonal terms of its reduced density matrix decay. The synchronization picture names what replaces it: the system *frequency-locks* (entrains) to the bulk reference, acquiring a definite *classical* phase relationship to Φ_bulk. These are two different quantities, not one conserved quantity — the quantum coherence is genuinely destroyed *in the reduced (local) state*, while classical phase-locking is established — so the transition is the loss of one and the onset of another, not a transfer. (Locally genuine, globally unitary: the local coherence is gone but the global evolution is the no-collapse, history-into- the-bulk scrambling of §3.1.) Because a single run carries one actual background configuration (§3.3), the result is a definite state rather than an improper mixture that merely looks classical. This supplies einselection [4] with a specific dynamical mechanism — phase-locking stability — for why pointer states survive: they are the states that can entrain to the bulk.

### 3.5 The reversible→irreversible boundary, formally: Re W and Im W

The two stages of §3.1 and the demonstrations of §3.2 have a compact formal counterpart. Integrating out the Dirac field in the presence of a background — an external field together with the conductor's boundary conditions — yields the vacuum persistence amplitude

$$\langle 0_{\text{out}}|0_{\text{in}}\rangle = e^{iW}, \qquad |\langle 0_{\text{out}}|0_{\text{in}}\rangle|^2 = e^{-2\,\mathrm{Im}\,W},$$

with $W$ the one-loop effective action. The probability that no real quantum is produced — that the in-state is *returned* — is $e^{-2\,\mathrm{Im}\,W}$; the probability that a real, on-shell quantum appears is $1 - e^{-2\,\mathrm{Im}\,W} \approx 2\,\mathrm{Im}\,W$ in the weak regime. The split of $W$ into real and imaginary parts is the reversible→irreversible boundary:

- $\mathrm{Re}\,W$ — **Stage 1, reversible.** The real, dispersive part carries the virtual-pair physics: vacuum polarization and charge screening, the image-charge and Casimir energetics near the conductor, and the resulting level shifts. These are *reactive* — they renormalize energies and dress the state, produce no on-shell quanta, and leave no record; they are even under time reversal. This is the catch-and-reverse window of §3.2: a fluctuation that borrows energy $\Delta E$ and repays it within $\hbar/\Delta E$. While only $\operatorname{Re} W$ operates, the evolution is unitary and any dephasing is refocusable.

- $\operatorname{Im} W$ — **Stage 2, irreversible.** The imaginary, absorptive part is nonzero *only* when a channel can go on-shell, and it *is* the rate at which real quanta appear — the decay rate of the in-state; it is odd under time reversal. For a constant field it is the Schwinger result,

$$\frac{2\operatorname{Im} W}{V_4} = \frac{e^2E^2}{4\pi^3}\sum_{n=1}^{\infty}\frac{1}{n^2}\,\exp\!\left(-\frac{n\pi E_{\text{crit}}}{E}\right), \qquad E_{\text{crit}} = \frac{m^2c^3}{e\hbar},$$

exponentially suppressed below the critical field. "Mostly reversible (Re), rarely real (Im)" is therefore exact and quantitative: the rare event is the exponentially small $\operatorname{Im} W$, and its threshold scale $E_{\text{crit}}$ is set by the mass.

Producing a real pair is necessary but not sufficient for a record; the record is *completed* when the on-shell quantum couples dissipatively to the bulk — the Adler/Stage-2 lock of §2.3. The electron relaxes into the conductor's electron sea at a capture rate $\Gamma_{\text{cap}} \sim K_{\text{eff}}$, the positron annihilates at $\Gamma_{\text{ann}} \propto n_e|\psi(0)|^2$ (overlap with the electron density), and the available final states are restricted by Pauli blocking through the Fermi factors $(1-f)$. A dense bulk does not *reverse* a real pair — a frequent misreading — it supplies the partners and dissipation that *complete* its irreversibility quickly.

The same Re/Im structure governs the general measurement, not only pair creation. For a system coupled to a detector bath the analogous object is the Feynman–Vernon influence functional $e^{i\Phi}$: $\operatorname{Re}\Phi$ renormalizes and shifts the system dynamics (reversible), while $\operatorname{Im}\Phi$ is the decoherence functional. For the linear, Gaussian coupling of §4.3 the latter is exactly the variance $\langle\delta^2\rangle/2$, so the visibility envelope $e^{-\langle\delta^2\rangle/2}$ is $e^{-\operatorname{Im}\Phi}$. The thermal source of §4.3 is a standard contribution to this $\operatorname{Im}\Phi$; the gravitational source of §6 would be a *second* contribution to the same envelope **only under the added postulate of §6.2**, and is absent without it — under standard QED the redshift cancels as a common-mode phase (Appendix C). (The vacuum persistence amplitude and the influence functional are distinct objects; we invoke them as two instances of one structure — reversible = Re, irreversible = Im — not as the same functional.)

**The Heisenberg cut, made precise.** The cut promised in §2.3 is the boundary between the regime in which only $\operatorname{Re} W$ (equivalently $\operatorname{Re}\Phi$) acts and the regime in which an imaginary part couples to an irreversible bath. It is *not* a fixed length, mass, or location; it is the joint condition

$$\operatorname{Im} W \ (\text{or } \operatorname{Im}\Phi) \neq 0 \quad \text{and} \quad \Gamma_{\text{cap}} > 0,$$

an on-shell channel open *and* the resulting quantum dissipatively coupled to the bulk. Either condition alone leaves the evolution reversible: an on-shell quantum with the bath removed ($\Gamma_{\text{cap}} \to 0$) is the engineered catch-and-reverse regime [14], where $\operatorname{Im} W$ may be nonzero yet no record forms; and a state dressed only by $\operatorname{Re} W$ never leaves the reversible window. Isolating or cooling the system *recedes* the cut; bringing a dense bulk to the interaction vertex *snaps* it there. The cut is thus the dynamical onset of absorptive coupling — exactly the closed→open transition of §2.

We are explicit about provenance. $\operatorname{Re} W$, $\operatorname{Im} W$, the Schwinger rate, the influence functional, and the Lindblad capture are standard results of quantum electrodynamics and open-system theory; the framework adopts them rather than deriving them. Its content here is the *identification* — Stage 1 = $\operatorname{Re} W$, Stage 2 = $\operatorname{Im} W$ gated by $\Gamma_{\text{cap}}$, the Heisenberg cut as the onset of absorptive bath coupling — which turns the reversible→irreversible boundary of §3.2 from a set of demonstrations into a condition with an explicit order parameter: the record probability $2\operatorname{Im} W$, gated by the bath lock $\Gamma_{\text{cap}}$.

Read this way, the detector is a *condensed-matter* system and the pointer is a collective degree of freedom of it: the record probability $2\operatorname{Im} W$ is its order parameter, and the Heisenberg cut is the driven–dissipative locking transition at which that order parameter becomes nonzero — a transition of the same family as quantum synchronization [10–13] and dissipative phase transitions more broadly, not a new postulate. The collectivity resides in the detector's bath coupling $\Gamma_{\text{cap}}$, consistent with the einselection of pointer states [4, 5] that §3.4 supplies dynamically.

### 3.6 The single-fermion cut: the dressed Dirac mass

The collective order parameter of §3.5 admits a one-particle *reduction* — the many-body content sits in the bath coupling $\Gamma_{\text{cap}}$ that is traced out below, not in $\psi$ — and writing it in that form collapses §2.2, §2.3, and §3.5 into a single statement about where the Dirac mass pole sits. Couple the fermion to the bulk and trace the bath out; the propagator acquires a self-energy $\Sigma$, and the pole — the physical mass — moves:

$$S(p) = \frac{i}{\not{p} - m - \Sigma(p)}, \qquad \Sigma(p) = \operatorname{Re}\Sigma - \tfrac{i}{2}\,\Gamma, \qquad m_{\text{eff}} = m + \operatorname{Re}\Sigma - \tfrac{i}{2}\Gamma.$$

That single complex number carries both stages:

- $\operatorname{Re}\Sigma$ — **Stage 1, reversible.** A shift of the mass, hence of the zitterbewegung beat frequency $2m_{\text{eff}}c^2/\hbar$. The fermion is *dressed* — its clock redialed — but undamped: the §2.2 regime of a real mass, with the chiral L↔R Bloch vector precessing on the surface of the sphere at conserved purity.
- $\operatorname{Im}\Sigma = -\Gamma/2$ — **Stage 2, irreversible.** The pole leaves the real axis. In the chiral/Bloch picture $m \to m - \frac{i}{2}\Gamma$ is non-Hermitian: the precession no longer circles the sphere but **spirals inward at rate** $\Gamma/2$ — exactly the inward spiral §2.2 identified as the signature of locking, now supplied by the bath. $\operatorname{Im}\Sigma$ *is* the strength of the Adler attractor of §2.3.

Two standard results pin $\Gamma$, and they meet the framework's two ingredients. The *rate* is the golden rule, $\Gamma(\omega) = 2\pi\, g_{\text{eff}}^2\, J(\omega)$: the decay rate is set by the bath spectral density $J$ *at the system frequency*, so $\operatorname{Im}\Sigma \neq 0$ only if the bath carries real modes resonant with the fermion — the on-shell condition of §3.5 as a condition on $J$. The *gate* is the Arnold tongue: linearizing the Adler equation $\dot{\phi} = \omega + K_{\text{eff}} \sin(\Phi_{\text{bulk}} - \phi)$ about the locked phase $\Delta_*$ gives $\dot{\delta\phi} = -K_{\text{eff}} \cos\Delta_*\, \delta\phi$, so the lock relaxation rate is

$$\Gamma = K_{\text{eff}} \cos\Delta_* \xrightarrow{\text{resonance}} K_{\text{eff}}, \qquad \text{a lock existing only inside } |\Delta\omega| \lesssim K_{\text{eff}}.$$

Hence $\operatorname{Im}\Sigma = -\frac{1}{2} K_{\text{eff}} \cos\Delta_*$: the bath coupling *is* the imaginary part of the dressed mass, and the Arnold-tongue capture condition of §3.1 and the golden-rule "$J(\omega) \neq 0$" are the same gate.

Everything then reduces to one inequality on a pole — the single-fermion form of the §3.5 joint condition:

$$\text{cut crossed} \iff \Gamma = -2\operatorname{Im}\Sigma > 0 \iff \underbrace{|\Delta\omega| \lesssim K_{\text{eff}}}_{\text{on-shell / Arnold tongue}} \;\wedge\; \underbrace{K_{\text{eff}} > 0}_{\text{coupled}}.$$

Closed or off-resonance, the pole stays on the real axis: real $m_{\text{eff}}$, reversible chiral precession (Stage 1). Opened and resonant, the pole moves off-axis: complex $m_{\text{eff}}$, the precession spirals into the lock (Stage 2). Because $\Sigma$ is causal, $\operatorname{Re}\Sigma$ and $\operatorname{Im}\Sigma$ are Kramers–Kronig partners — the reversible redialing of the clock and its irreversible damping are two faces of one analytic object. The **Heisenberg cut is the dressed Dirac mass pole leaving the real axis.**

We are explicit about provenance and scope, as in §3.5. The self-energy and pole structure, the golden rule, Kramers–Kronig, and the Adler linearization are standard; the framework's content is the *identification* $\operatorname{Im}\Sigma = -\frac{1}{2} K_{\text{eff}} \cos\Delta_*$ — that the bath coupling entering as $\operatorname{Im}\Sigma$ is the Adler lock rate — which we state as an identification, not a derivation. We also keep $\Sigma$ to its dominant scalar (mass-dressing) channel; the full self-energy is a $4\times4$ matrix with additional Lorentz structure, and we do not claim that general decomposition here; §3.7 makes that restriction precise, and shows it runs the wrong way for the electromagnetic coupling.

### 3.7 What the measurement coupling can and cannot supply

The equivariance step that §8 will ask of Stage 2 — that the lock carry the prepared $|\cdot|^2$ weight intact to the outcome — presupposes that the Stage-2 coupling is *non-demolition* for the measured observable: the dissipator must preserve the populations of the measured projector while scrambling its phase. It is worth stating which observable the framework's own coupling actually treats this way, because the answer is not the one the chiral picture invites.

In the chiral reduction the interference channel is the scalar bilinear $\bar{\psi}\psi = \sigma_x$ — the mass channel — whose two pointer phases $0, \pi$ are the binary of §3.2. A non-demolition measurement of it requires the bath to couple *through* $\sigma_x$. But the electromagnetic interaction couples through the **vector current** $\bar{\psi}\gamma^\mu\psi$, and the chiral classification of Dirac bilinears is exact: vector and axial currents are chirality-*preserving* ($\bar{\psi}_L\Gamma\psi_L$), while the scalar, pseudoscalar, and tensor bilinears are chirality-*flipping* ($\bar{\psi}_L\Gamma\psi_R$). The gauge field therefore couples only to the charge density ($j^0 = \mathbb{1}$) and the chirality-diagonal current ($\sigma_z$); it cannot generate the $L \leftrightarrow R$ mixing $\bar{\psi}\psi$. (The magnetic-moment, or Pauli, term is chirality-flipping, but it acts on *spin* — coupling spin to $\mathbf{B}$, as in Stern–Gerlach — not on the chiral scalar.) Equivalently at the self-energy level: in $\Sigma = \Sigma_V(p^2)\not{p} + \Sigma_S(p^2)m$ the scalar part $\Sigma_S \propto m$ is protected by chiral symmetry ($\Sigma_S \to 0$ as $m \to 0$), so the absorptive piece $\operatorname{Im}\Sigma_S$ — the mass-channel attractor invoked in §3.6 — is the *chirally suppressed* one, while the unsuppressed dissipation lives in the vector channel $\operatorname{Im}\Sigma_V$.

The consequence is concrete. Carried through to a basin split, the $\sigma_z$ coupling reproduces no Born weights: for the rest-frame mass Hamiltonian it is transverse and drives *relaxation* toward the $\sigma_x$ ground state (one pointer at $T = 0$, a Boltzmann mixture at $T > 0$); away from rest it is longitudinal and *dephases* in the chirality basis. In every case the outcome probability becomes independent of the prepared phase — the input weight is erased, the opposite of equivariance. The electromagnetic coupling is thus non-demolition for the charge/position/chirality pointer basis (ordinary which-path einselection, §3.4, which we keep) but *not* for the chiral interference channel. The two-basin $\sigma_x$ structure is, on this analysis, a feature of the Stage-1 internal clock — the zitterbewegung beat of §2.4 and [26] — rather than the pointer basis the environment selects.

The single coupling that *is* non-demolition for $\bar{\psi}\psi$ is one that couples to the scalar/mass density directly — a Yukawa scalar, or gravity, since $\bar{\psi}\psi$ is (on the equations of motion) the trace of the stress–energy tensor, $T^\mu{}_\mu = m\bar{\psi}\psi$. This is the rigorous sense in which the Born-compatible coupling is gravitational, and it carries two costs that §7.4 and §8 take up: the Penrose self-energy timescale $\hbar/E_G$ is negligible for microscopic systems ($\sim 10^{16}$–$10^{24}$ s for an atom or electron), becoming fast only at macroscopic pointer masses; and the objective-collapse programs that use this same operator do not *derive* the Born weights either, but normalize them in. We therefore do not claim that the framework's coupling supplies the measured channel equivariance needs: the electromagnetic interaction supplies a different one, and the coupling that would supply it is gravitational, rate-suppressed, and no better off than Penrose–Diósi on the weights themselves. The Born measure (§8) is, on this analysis, uniformly open across the synchronization and gravitational-reduction readings alike.

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## 4. The bulk reference

### 4.1 What the bulk is: a faint thermal bias, not a coherent macrostate

It is essential not to overstate the bulk's coherence. The reference is **not** a macroscopically entangled or coherent quantum state. The cleanest quantitative handle is magnetic resonance: the equilibrium nuclear polarization that defines the bulk reference axis in MRI is P $\approx$ $\hbar\gamma$B$_0$/2k_BT $\approx$ 5 parts per million at 1.5 T and body temperature. The bulk is ~99.999% thermal, disordered, and mixed; the "reference" is a faint statistical bias on top of a hot reservoir. That faint bias is nonetheless sufficient to image the proton distribution of the body — existence proof that a measurement reference need not be a coherent macrostate.

We therefore drop the language of an "entangled bulk." The reference is a *phase-synchronized bias*, maintained by whatever interaction dominates locally.

Two attributes of that bias must be held apart, because temperature treats them differently. Write the reference as a complex order parameter $r\,e^{i\psi}$: a *magnitude* $r$ — how large the bias is, the parts-per-million above — and a *phase* $\psi$ — the reference *axis*, the direction the bias points, fixed by the dominant interaction (the direction of $B_0$ in MRI). Temperature suppresses the magnitude but not the axis. The ~99.999% thermal disorder is exactly what holds $r$ down to ppm; it does **not** erase $\psi$, which is pinned by the deterministic locking. The bulk is therefore *faint but oriented* — small amplitude, definite axis — and these are independent properties, a point §4.3 makes dynamical. That the axis $\psi$ survives what temperature destroys is not incidental: the axis is the coherent content any reference must carry — a vanishing amplitude $r$ can still serve, but a bath with no definite axis is no reference at all (§4.2).

### 4.2 Absolute phase is gauge; only differences are physical

A natural objection is that the bulk's phase Φ_bulk cannot be measured — any probe prepared with a known phase synchronizes to the bulk on contact, so its absolute value is hidden by construction. This is not fatal; it is the same status that global phase, absolute gravitational potential, and absolute velocity have throughout physics. The discipline it imposes is that **every physical claim must reduce to a phase difference.** Accordingly, the framework's entire observable content involving Φ_bulk lives in differences between regions — and there is exactly one such observable difference, the gravitational case of §6. Anything said about Φ_bulk that does not reduce to a measurable difference is interpretation, and we label it as such.

One distinction must stay sharp, because the revision turns on it: the unobservability of the *absolute U(1) phase* is a gauge fact — only phase differences are physical — and it is independent of the separate question of a preferred Lorentz *frame*. The framework keeps the former and, as §8 makes explicit, does **not** thereby claim the latter away: the bulk defines a physical rest frame even though its absolute phase is gauge. "Absolute phase is hidden" is not the same statement as "there is no preferred frame," and we do not use the first to imply the second.

A familiar optical system makes this discipline concrete: holography. A hologram fixes on the plate the interference of an *object* wavefront with a coherent *reference* beam — that is, their *phase difference* (the fringe pattern), never either phase alone. Add a common global phase to object and reference alike and the fringes are unmoved — the gauge fact, "absolute phase is hidden"; advance one *relative* to the other and the fringes shift by a readable amount — the physical difference. (Phase-shifting holography reads out precisely this differential; an off-axis reference contributes only a fixed common carrier that drops out of the reconstructed object, the optical echo of an unobservable uniform background.) One feature of the analogy is load-bearing rather than decorative: **a hologram cannot be recorded in incoherent light.** Without a phase-stable reference there is no stationary fringe — only a washed-out average. This is the optical form of the framework's central requirement, that a measurement reference be phase-coherent — *one cannot lock to noise* — and it is why Φ_bulk cannot be identified with the phase-random zero-point vacuum ($\langle E\rangle = 0$), which belongs instead to the stochastic forcing $\eta$ of §4.3. We give this as analogy, and label it as such; the frame claim itself rests not on the analogy but on the single global difference of §6.

The coherence requirement also separates three roles the word "vacuum" otherwise blurs. (i) The condensate that locks L↔R and dresses the Dirac mass (§2, §3.6) is a coherent vacuum expectation value — a definite-valued reference. (ii) The measurement reference Φ_bulk of this section is likewise coherent and oriented, but its frame-bearing *selection* is the open problem of §8, not a settled value. (iii) The zero-point fluctuations — phase-random, $\langle E\rangle = 0$ — are not a reference at all but the decohering noise of §3.5 and §4.3. Holography forbids collapsing (iii) into (i)–(ii): incoherent light records nothing. The framework's outstanding obligation is thus confined to (ii) — supplying a coherent, frame-bearing Φ_bulk — and §8 carries it as such.

### 4.3 Magnitude and rate are independent

The smallness of the reference (§4.1) does not weaken the mechanism. The equilibration *rate* — how fast a perturbed system re-locks to the bulk — is set by the coupling to the bath (in MRI, T1 and T2), and is independent of the order-parameter *magnitude* (the ppm polarization). A faint reference can drive fast, sharp, definite relaxation. Measurement definiteness rides on the rate, not on the order-parameter size.

This separation has a dynamical companion, and it is where temperature enters as an explicit *randomization term in the phase synchronization itself.* Write the deviation of a locking system from the bulk phase as $\delta$; near the lock it obeys

$$\dot{\delta} = -K\delta + \eta(t), \qquad \langle \eta(t)\eta(t') \rangle = 2D\,\delta(t - t'), \qquad D \propto k_B T,$$

a deterministic restoring force of strength $K$ that pins the axis, plus a stochastic thermal forcing $\eta$. The steady state is Gaussian with variance $\langle \delta^2 \rangle = D/K$, and the coherence amplitude of the lock is $r = e^{-\langle \delta^2 \rangle/2}$. This makes the §4.1 amplitude–axis split precise as a statement about *moments.* Temperature acts through one of them only: it widens the phase *variance* (the second moment) and thereby damps the amplitude $r$, while the *mean* phase — the axis $\psi$ (the first moment) — is held by the restoring force and does not move. "Temperature randomizes the phase" means exactly this: it inflates $\langle \delta^2 \rangle$, lowering the amplitude, not rotating the axis. The amplitude does depend on the *ratio* $D/K$, so it is tied to the lock rate; the equilibrium polarization of §4.1 and the axis are the quantities that vary independently of it. This thermal half of the envelope is not just a model: it is the measured behaviour of matter-wave interference. Heating $C_{70}$ fullerenes until they emit thermal photons reduces the fringe visibility in quantitative agreement with decoherence theory [28] — temperature damps the amplitude, not the fringe positions, exactly the amplitude-not-axis statement above. The same envelope $e^{-\langle \delta^2 \rangle/2}$ — the influence-functional form $e^{-\mathrm{Im}\,\Phi}$ of §3.5 — reappears in §6, where the variance is supplied not by temperature but by a far smaller gravitational phase difference: one slot, two candidate sources — the thermal one measured and standard, the gravitational one contingent on the postulate of §6.2 and absent without it.

### 4.4 The role of gravity, stated honestly

In ordinary matter the interaction that maintains the bulk reference is electromagnetic — $B_0$ in MRI, lattice and Coulomb binding in a solid-state detector, supersaturation metastability in a cloud chamber. Gravity is negligible there by ~30 orders of magnitude, and we make no claim that gravity maintains everyday detector coherence. In particular, we do **not** assert a gravitational coherence rate of the form Γ_grav ~ $GM^2/(\hbar\Delta z)$: summing $N^2/2$ unnormalized pairwise gravitational couplings does not yield a physical relaxation rate (the mean-field coupling must be normalized intensively), and the resulting figure is a dimensional artifact, not a derived rate.

Gravity enters in exactly one place: where the electromagnetic relaxation channel has been deliberately engineered away (the cold, isolated, mesoscopic regime of Penrose–Diósi tests), and, separately, through the ordinary gravitational **redshift** of the electromagnetically-maintained reference clocks at different potentials — which is the basis of the prediction in §6. The first is the classicalization regime Penrose's objective reduction [15] aims to describe; the second is standard general relativity applied to the detector references, not a new gravitational coherence mechanism.

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## 5. Three regimes of capture

A single mechanism — resonant capture (Stage 1) followed by bulk equilibration (Stage 2) — spans measurements that look very different, distinguished only by the *capture strength per event* and the *number of events.* The unifying axis is how deep inside the Arnold-tongue locking range each capture sits.

| | Stage-1 capture | Reversible window | Stage-2 irreversibility | Reference maintained by |
|---|---|---|---|---|
| **Bell/photon detector** | photon resonantly absorbed by a bound electron | brief (resonance fluorescence) | electronic **amplification** (avalanche, PMT gain) | EM lattice binding |

| | Stage-1 capture | Reversible window | Stage-2 irreversibility | Reference maintained by |
|---|---|---|---|---|
| **NMR / MRI** | RF pulse tips M coherently | **T2* — refocusable by spin echo** | T1/T2 relaxation to lattice | **$B_0$ (applied EM field)** |
| **Cloud / bubble chamber** | a *sequence* of weak partial ionizations | partial, per event | stepwise droplet/bubble nucleation | EM (metastable supersaturation) |

Three readings of the same structure:

- **Bell detector — the strong/projective limit.** One event saturates the locking range; the irreversibility is the amplification, not the bare excitation (a single excited electron can re-emit; the avalanche cannot be undone). One quantum, one Stage 1, one Stage 2, one click.
- **MRI — isolated, deliberately-triggered Stage 2.** Here $B_0$ sets the reference axis and Larmor frequency $\omega_0 = \gamma B_0$ (the role of "Φ_bulk"); the *gradient* coils are not the reference but the spatial encoder, the direct analog of the Stern–Gerlach gradient that converts internal phase into a position/frequency label. The RF pulse is the coherent capture; relaxation is Stage 2, watched directly. Every free-induction decay is, in framework terms, a Stage-2 emission measurement.
- **Cloud chamber — the weak/repeated limit.** The charged particle undergoes a *sequence* of distinct vertices, each a weak partial capture (removing tens of eV from a MeV–GeV particle) seeding a local Stage-2 nucleation. It is not one drawn-out Stage 2 but many partial (Stage 1 + Stage 2) pairs. The collinearity of the track (Mott 1929 [16]) measures the per-event partiality: each capture only partially sharpens the momentum, so the next fires preferentially along the line of flight. dE/dx is the per-event coupling strength integrated over the trajectory.

The same incoming fermion leaves a single click in a silicon pixel and a millimeter trail in a chamber because the two architectures exercise opposite ends of the capture-strength axis — projective saturation versus near-threshold partial capture — not because the underlying vertex differs.

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## 6. A candidate prediction: a linewidth-dependent gravitational Bell test, and the postulate it requires

### 6.1 Statement of the candidate effect

The framework *suggests* an observable departure from standard quantum optics — but, as we show in §6.2 and derive in Appendix C, the effect does **not** follow from the framework together with standard quantum electrodynamics. It requires an additional, non-standard postulate. We state the candidate effect here and isolate the postulate it rests on in §6.2; the experiment is best read as a falsification test *of that postulate*, not as a clean consequence of the framework.

Two detectors A and B in a photonic Bell test sit at gravitational potentials Φ_A ≠ Φ_B. Their electromagnetically-maintained reference clocks are redshifted relative to each other by the standard factor $\Delta\Phi/c^2$. *If* the local reference couples into the polarization measurement (the postulate of §6.2), then over a measurement the relative reference phase accumulates

$$\delta\phi_{\mathrm{grav}} = \omega\,\frac{\Delta\Phi}{c^2}\,\tau_{\mathrm{coh}} = \frac{\omega\,\Delta\Phi}{c^2\,\Delta\nu},$$

where $\tau$_coh = $1/\Delta\nu$ is set by the photon **linewidth**, independent of central energy. The *functional form* −cos(2(a−b)) is set by the polarizer geometry (a reversible basis selection that precedes Stage-1 capture) and is gravitationally invariant. The *contrast* would then degrade as

$$\mathrm{CHSH}(\Delta\nu) = 2\sqrt{2}\,\exp\left(-\frac{\delta\phi_{\mathrm{grav}}^2}{2}\right)$$

(Gaussian line). Standard QM/QED predicts CHSH independent of linewidth at fixed entanglement fidelity; the candidate effect is a linewidth-dependent degradation in a gravitational gradient. Whether it exists at all hinges on §6.2.

### 6.2 The postulate the effect requires, and why it does not follow from standard QED

**Magnitude, conditional on the coupling.** Were the local reference to couple into the projection, the redshift would enter as a *steady, deterministic* frequency offset, not a stochastic per-period kick; a steady offset accumulates phase linearly in time, so $\delta\varphi \propto \tau$_coh $= 1/\Delta\nu$. The contrary "per-optical-period floor" reading, which would make any effect negligible, is incorrect for a steady offset. This fixes the *scaling* of the candidate effect — but says nothing about whether the coupling exists.

**It does not follow from standard QED (Appendix C).** We attempted to derive the required coupling from the standard light–matter Hamiltonian. The absorption matrix element factorizes into a polarization projection (carrying the analyzer angle) times a temporal/energy factor (carrying the local-clock phase, redshift included). These multiply; they do not mix. The clock phase enters the detection probability only through the modulus of the temporal factor, $|\mathrm{T}|^2$, which is independent of the analyzer angle — so the redshift is a common-mode phase that cancels in the normalized correlation. Standard QED therefore predicts CHSH independent of linewidth: the null.

**For the effect to exist, the H and V branches must carry different local-clock phases** — the local temporal reference must couple to the *polarization projection itself.* We checked every apparatus element that could plausibly supply this, and none does:

| Element | Dissipative? | Polarization-dependent phase? | Carries A–B redshift into the correlation? | Why it fails |
|---|---|---|---|---|
| Detector electron | yes | no | no | H and V absorbed at the same potential Φ_A → common local phase |
| Polarizing beam splitter | no (unitary) | yes (birefringence) | no | reversible → a recalibratable, linewidth-independent phase, not a decoherence envelope |
| Absorbing polarizer | yes, on the *rejected* photon | no | no | post-projection the transmitted photon retains no H/V memory; same-potential null |
| Path deflection / crystal recoil | normally negligible / reversible | yes (recoil) | no | Stern–Gerlach-like and reversible; any recorded recoil references the same crystal at Φ_A for both branches → common phase |

The four failures share one root cause: at a single detector both Bell branches reference the *same* local potential, and every genuinely dissipative step occurs *after* polarization projection, when the photon no longer carries which-branch information. The A–B redshift can enter the correlation only through a coherent, pre-projection, polarization-*differential* phase — and the sole element that fits (birefringence) is unitary, recalibratable, and linewidth-independent.

**The required postulate, stated explicitly.** The candidate effect therefore rests on an additional term, schematically

$$H' \sim g\,(\mathbf{d} \cdot \hat{\mathbf{H}})\,\phi_{\text{bulk}},$$

coupling the polarization projection directly to the local background phase. This term is (a) absent from standard QED; (b) non-covariant — it singles out the analyzer axis and so breaks the rotational symmetry of the dipole interaction; and (c) tied to the absolute bulk phase, and hence to the physical preferred frame the framework openly carries (§8), whose Lorentz-covariant formulation remains open.

**What the experiment tests.** Because the effect does not follow from the framework plus standard optics, the experiment is a clean falsification test *of the postulate* $H'$, not of the framework's core. A null result (CHSH flat versus linewidth) confirms standard QED and disconfirms $H'$; a degradation scaling as $1/\Delta\nu^2$, present only with an altitude split, would be evidence for it. We do not claim the effect is a derived consequence of MCI — we have shown it is not — and we present it as the one place the framework *could* be made to touch experiment, contingent on a postulate we cannot currently derive.

Two consistency notes survive the demotion. First, even under $H'$ the clock phase enters only as a multiplicative visibility envelope (a contribution to the $\operatorname{Im}\Phi$ decoherence functional of §3.5, present only under $H'$); it does **not** generate the correlation. Standard QM supplies both −cos2(a−b) and the $2\sqrt{2}$ ceiling — the clock-as-hidden-variable model alone is sub-classical (Appendix A) — so the framework never double-counts the clock as the source of the correlation. Second, the established gravitational-time-dilation visibility effects [24, 25] do not apply here: they require a real internal clock in a spatial superposition straddling both potentials, which the Bell photons lack — which is precisely why an extra postulate is needed rather than an existing mechanism.

### 6.3 Consistency with existing satellite tests

Existing long-baseline Bell tests do not already constrain the postulate of §6.2. The Micius satellite experiments and ground tests use broad-linewidth spontaneous parametric down-conversion (SPDC) photons ($\Delta\nu$ of order GHz–THz); for those, $\delta\varphi$_grav is of order $10^{-7}$ or smaller even over 500 km altitude differences, far below resolvability. The candidate effect is specifically *narrow-linewidth*: it would require cavity-filtered photons ($\Delta\nu$ ~ kHz) to bring $\delta\varphi$_grav to order unity at km-scale altitude differences. This is why the signature has never been tested — no Bell test has combined narrow linewidth with a gravitational gradient.

### 6.4 Parameter space and falsification

The discriminating measurement is a joint linewidth × altitude scan with cavity-filtered narrow-linewidth entangled photons. Photon central energy is not the discriminator; linewidth $\Delta\nu$ and altitude split $\Delta\Phi$ are. Order-of-magnitude: $\delta\varphi$_grav ~ 1 requires roughly $\omega(\text{gh/c}^2)/\Delta\nu$ ~ 1, i.e. for optical $\omega$ ~ $4\times10^{15}$ rad/s and a few-km altitude split (gh/c$^2$ ~ $3\times10^{-13}$), a linewidth of order kHz. A calibrated scan at fixed entanglement fidelity that holds CHSH flat across linewidth would disconfirm the postulate $H'$ of §6.2 (and confirm standard QED); a degradation scaling as $1/\Delta\nu^2$, present only with an altitude split, would be evidence for it.

## 7. Relation to other interpretations

### 7.1 An honest placement: the single-world, $\psi$-ontic, nonlocal family

MCI takes $\psi$ to be a real physical field ($\psi$-ontic, in the sense of Harrigan and Spekkens [17], consistent with PBR [18]) and adds a beable: the actual configuration of the background/vacuum field at each event, whose value selects the outcome. Its proper category is therefore the **single-world, $\psi$-ontic, nonlocal hidden-variable** family — alongside Bohmian mechanics [2, 3] (beable: particle position) and Nelson's stochastic mechanics [19] (beable: a stochastic background). This is the framework's real intellectual home, and naming it honestly is more durable than positioning MCI as a slayer of any rival.

The $\psi$-ontic reading is not a bare metaphysical preference; it is the stance under which several long-established results of single-particle Dirac theory read most naturally. Ohanian [38] recovers the electron's spin and magnetic moment as the angular momentum of the *circulating energy flow* and the moment of the *circulating charge flow* of the Dirac field — observables obtained by treating the field as a real, spatially-extended distribution rather than a probability amplitude over a point. The Gordon decomposition [39] makes the structure explicit: the conserved Dirac current splits into a convection current and a spin magnetization current, which in the exact Dirac–Coulomb solution of hydrogen [40, 41] are the real circulating currents of a stable, spatially-distributed bound state — the orbital current carrying the orbital moment, the magnetization current carrying the spin moment with g = 2 — whose interference reproduces the fine structure, and whose internal circulation sits at the Compton/zitterbewegung scale of §2.4. None of this *proves* field realism — each result also admits an operational reading, and the integrated spin and moment are notably independent of the wave-packet envelope, so the support is for an *extended real field*, not a particle of definite radius. But the field-realist interpretation is the one under which spin, the magnetic moment, and the hydrogen spectrum are properties of a single real, extended object; we take that consonance as independent motivation for the $\psi$-ontic commitment, not as a derivation of it.

### 7.2 Against Many-Worlds: a trade, not a conquest

The contrast with Everett [1] is a *trade of costs*, and we state it as such. The moment one insists on a single world with definite outcomes that reproduces quantum statistics, Bell's theorem forces nonlocality. MWI escapes Bell only by denying definite single outcomes — it buys **locality** at the price of **many worlds**. MCI makes the opposite trade: it buys a **single world** at the price of **nonlocality** (in the Bell sense — not signaling; §7.5). No interpretation pays neither price. So MCI does not *abolish* the nonlocality that troubled Einstein; it relocates the cost from branching worlds to nonlocal beables, as Bohm does. We accordingly describe MCI as *a single-world alternative to Many-Worlds*, not a replacement that escapes its difficulties for free.

### 7.3 Against Bohm and Nelson: what MCI still owes

Within its own family MCI is currently the least complete member, and honesty requires saying so. Bohm has an explicit dynamical law (the guidance equation) and a clean account of the Born measure (quantum equilibrium, $|\psi|^2$). MCI now adopts the first — the coherent guiding law is the Madelung velocity $\mathbf{v} = \nabla\varphi/m$ of §2.2, read ontologically (§2.5) — but still owes the rest: no specified equation for *how* the background configuration selects the basin (the *dynamical* locus of its nonlocality, though the *ontological* locus is identified in §7.5), and an incomplete Born account. Reading $|\psi|^2$ as the energy density of a real field is true but, for the equal-energy channels the framework treats, coincides with the occupation fraction by normalization and so explains nothing about outcome statistics; and a genuinely unbiased background would, without further input, weight basins by their measure (tending to equal odds) rather than by $|\alpha|^2$ — the "why squared" gap that contested branch-counting derivations also face.

This recasting also fixes what a superposition *is*, and the point is worth making because the careless version invites a hidden-variable misreading. What is definite is the field: a system "in superposition" is a single definite $\psi$-configuration that merely fails to be an eigenstate of the observable one chooses — superposition is a relation between the definite field and a chosen reference basis, not an indefiniteness in the field. What is *not* a property of $\psi$ is the outcome, which is co-produced at Stage 2 by $\psi$ together with the per-run background configuration (§3.3) and fixed only when that background is sampled. This is precisely Bohm's structure — a definite $\psi$ with a contextual outcome — with the beable changed from particle position to background field, so that "definite" never attaches to the outcome and no local outcome-predetermining variable is smuggled in (Appendix A). The reading earns its place: an observable's result is background-sensitive exactly when $\psi$ is not its eigenstate, so "superposition with respect to A" *is* the condition that the Stage-2 selection depends on the background, and the Born weight $|\langle \mathrm{a}|\psi\rangle|^2$ is, on this reading, the measure of those backgrounds that select basin a. Why that measure takes the squared-overlap value rather than another is then the "why squared" gap restated — the framework's counterpart of Bohm's quantum-equilibrium $|\psi|^2$ over position — and is carried as an open problem in §8.

Against these debts stand three distinctions, and they are sharper than "a more physically-motivated beable."

First, **the guiding field is physicalized, not posited.** Bohm's quantum potential $Q = -\hbar^2\nabla^2 R/2mR$ is read off the amplitude of $\psi$ after the fact; it has no source of its own and no dynamics independent of the wavefunction it is extracted from. In MCI the analogous guidance is *inherited from the vacuum the particle traverses*: minimal coupling is carried not by the bare particle wavefunction but by the vacuum's virtual-pair fluctuations, to which the particle phase-locks, so the particle's phase is the vacuum phase it tracks [26]. We are precise about which half of Bohm's amplitude–phase split this captures. The directly physicalized quantity is the *phase* — the Hamilton–Jacobi $S$, exhibited cleanly as the Aharonov–Bohm phase $e\Phi/\hbar$ recovered from zero-frequency locking [26] — not literally the amplitude-side $Q$. The amplitude side enters through the *same* coupling's fluctuations: the variance $\langle\delta^2\rangle$ of the residual phase mismatch about the locked solution damps fringe contrast, and that loss of $R$-coherence is the decoherence (amplitude) physics. Where Bohm carries two formally separate objects — a guidance equation acting on $S$ and a quantum potential built from $R$ — MCI traces both to one mechanism, the particle's locking to the fluctuating vacuum: the deterministic lock supplies the guidance, its fluctuation variance supplies the amplitude/decoherence side. We present this as a structural correspondence, not an identity $\langle\delta^2\rangle = Q$.

Second, **it is relativistic from the start.** Textbook Bohm begins from the Schrödinger equation and must then have a preferred foliation grafted on to make its nonlocal dynamics covariant [27] — the standard charge that Bohmian mechanics is non-relativistic. MCI's Stage-1 dynamics is instead built from Lorentz-covariant Weyl/Dirac objects: the off-diagonal chiral mass coupling $L \leftrightarrow R$, with zitterbewegung as the $L$–$R$ beat (§2). Covariance is therefore present in the pilot dynamics by construction rather than added afterward. (Single-particle relativistic Bohm already exists in the Bohm–Dirac current-guidance form [27]; MCI's added content is the Stage-2 synchronization that supplies definite outcomes and the vacuum beable, not the relativistic single-particle law itself.)

Third — the sharpest contrast — **Bohm's preferred frame is undetectable; MCI's is physical.** The foliation Bohm needs is, by construction, empirically inaccessible. MCI's nonlocality is instead anchored to a *physical* preferred frame, the bulk/vacuum rest frame (§8), whose effects are in principle observable. We take the conservative reading: ordinary matter couples to that frame, beyond the electromagnetic Stage-2 locking, only gravitationally, so the residual anisotropy in everyday particle physics is gravitationally weak (§8). The companion analysis [26] develops the stronger, *kinematic* coupling to the same frame — a velocity-dependent vacuum-locking envelope — and proposes a fringe-visibility-versus-$\gamma$ scan as its direct test; that experiment is what would decide whether Bohm's hidden foliation has, in this framework, become a measurable physical frame.

The distinctive debt remains the two items above. The distinctive appeal is now stronger than a slogan about beables: the beable, the guidance, and the locus of nonlocality are one physical thing — the vacuum the particle is locked to — rather than three independent postulates.

### 7.4 Penrose–Diósi

Both this framework and Penrose–Diósi [15, 20] cast gravity as an agent of classicalization. We localize that role precisely: gravity acts at the electromagnetically-suppressed regime (Stage 2 in cold, isolated, mesoscopic systems), never on the basis selection that builds the correlation. Penrose's objective reduction is a candidate description of exactly that regime; the two proposals single out the same physical step (bulk re-establishment of coherence) and differ in dynamical reading — threshold instability versus continuous rephasing.

Two refinements sharpen that contrast and locate where the readings part company. First, the divergence runs deeper than threshold-versus-continuous: Penrose–Diósi is a *collapse* theory — it modifies Schrödinger evolution and the superposition is objectively destroyed — whereas MCI is no-collapse, the global evolution staying unitary with the coherence scrambled into the bulk (§3.1). Second, the two mechanisms ask opposite things of the environment: MCI's Stage-2 lock is *dissipative* and needs a bath to carry the phase mismatch into a continuum (§3.5), while Penrose's reduction is *intrinsic* and needs none, being a property the superposed mass distribution carries on its own. In the deep Penrose regime — cold, isolated, mesoscopic, the electromagnetic and thermal channels deliberately removed — that distinction is decisive: the bath MCI would dissipate into has been engineered away, and we do not assert a gravitational dissipation channel to take its place (§4.4). We flag a tension rather than hide it: §3.7 finds that the only coupling able to perform a non-demolition measurement of the chiral channel — and so to complete the Born account — is a coupling to the scalar mass density $\bar{\psi}\psi$, i.e. exactly the gravitational channel this paragraph declines to assert; completing Born would thus narrow the operator-level gap to Penrose–Diósi to the bare collapse-versus-no-collapse distinction. The honest statement is therefore not that MCI subsumes Penrose but that the gravity-dominated regime is exactly where MCI's dissipative mechanism runs out of bath while Penrose's intrinsic one does not — which makes that regime the arbiter between them: a continuous, no-collapse reading (a gravitational coupling that damps coherence smoothly, *below* any Penrose threshold) and an objective collapse predict different things there, and the companion analysis [26] pursues the continuous branch as a gravitational reduction of the vacuum-locking rate.

### 7.5 Bell, no-signaling, and the "unhidden variable"

We restate plainly: the framework is **not** superdeterministic. Detector settings are free; the shared bulk environment is a *public, setting-independent* variable that Bell's theorem permits and that does nothing to relax any Bell assumption. There is no faster-than-light signaling — Bell nonlocality is not signaling, and MCI, reproducing standard quantum statistics, inherits the no-signaling theorem exactly as Bohm and ordinary QM do. The bulk reference is an "unhidden" variable only in the benign sense: it makes explicit the phase coherence that realistic descriptions of macroscopic detectors already assume. It is not a Bell-relevant hidden variable, and it is not a route around nonlocality.

It is worth being precise about *which* bulk this denies and which it does not, because the same distinction locates the framework's nonlocality. Two physically different objects travel under the word "bulk," and only one is the setting-independent common cause just excluded:

- **The local thermal reference** at each detector — the faint, ppm bias of §4 ($B_0$ in MRI, lattice and Coulomb binding in a solid-state detector). This performs the local Stage-2 lock, the registration, and — through its coherence amplitude $r = e^{-\langle\delta^2\rangle/2}$ (§4.3) — the visibility of the outcome. It is established in the common past of both wings and is independent of the analyzer settings, so it is exactly a Bell common-cause variable; by the theorem (and quantitatively by Appendix A, CHSH $\leq \sqrt{2}$) it *cannot* be the source of the correlations. Its coherence magnitude is irrelevant to this — a more coherent common cause is still bounded by CHSH $\leq 2$ — so the smallness of the ppm bias is not the obstacle and additional coherence is not the cure. Cooling a detector lowers $D \propto k_B T$ (§4.3), raises $r$, and thereby sharpens the *visibility* of an entanglement-sourced violation; it does not, and cannot, convert this local reference into a nonlocal channel.
- **The extended, non-separable field configuration** carrying the entangled two-particle mode across both wings. This is not a common-cause variable at all: it is a genuinely non-separable (entangled) configuration of the $\psi$-ontic background field, and non-separability is precisely the property a local common cause lacks. *This* is where the framework's nonlocality resides. Ascribing it here is not a new postulate but a restatement of the $\psi$-ontic commitment of §7.1 — the wavefunction is a real field, and an entangled pair is one extended, non-factorizable configuration of it spanning A and B.

Naming the locus this way settles, at the level of *ontology*, the gap §7.3 flagged ("no explicit locus for its nonlocality"): the nonlocal element is the non-separable field configuration, not the local thermal bulk, which does only registration and visibility. It does **not** yet settle the *dynamics* — an explicit law for how that non-separable configuration updates across a spacelike interval (the selection law of §8), together with a proof that the update preserves no-signaling. The preferred frame the framework already carries (§8) supplies the simultaneity such a law needs; the no-signaling proof does not follow for free once the dynamics is written independently of the quantum statistics it must reproduce, and securing it is part of the debt of §8.

### 7.6 Relational clocks and real-clock decoherence

The framework's title and its central image — that every particle carries an internal phase clock, and that measurement is the mutual synchronization of such clocks — place it in deliberate proximity to the relational-time tradition, and we state the relationship precisely rather than borrow the vocabulary loosely.

In the **Page–Wootters** mechanism [29, 30] and its modern operational formulations [31–33], time is not a background parameter but a correlation: the global state is stationary, and dynamics is the conditional dependence of a "system" on the reading of a "clock." MCI uses *clock* in a related but distinct sense. Its clocks are physical phase degrees of freedom — the chiral/Compton phase whose $\pm E$ beat is the Zitterbewegung (§2.4) — and the operation that matters is not the *definition* of a time parameter from a clock reading but the *dissipative phase-locking* of many such clocks to a common reference (§3). MCI does not derive relational time; it takes many physical clocks as given and asks what their entrainment does at a measurement. The two pictures are compatible — a Page–Wootters clock is exactly the kind of physical phase MCI synchronizes — and the quantum-reference-frame treatment of *interacting* clocks [32, 33] is the natural language in which the Stage-1 coupling of a system clock to a detector clock would be made rigorous. We note this as a possible formal home for the framework's kinematics, not as a result we have established.

Two relational programs bear more directly on the framework's claims. First, the **thermal time hypothesis** of Connes and Rovelli [34]: for a system in a thermal (KMS) state, the modular flow of that state plays the role of physical time, so a thermal reservoir does not merely *sit at* a temperature but *defines a preferred flow*. This is suggestive for §4, where a faint, overwhelmingly thermal bulk supplies the measurement reference: thermal time offers a candidate first-principles account of why a hot, disordered reservoir can nonetheless fix a definite reference *axis* (the §4.1 amplitude/axis split), and is, we think, the most promising existing framework in which the bulk reference could be grounded dynamically. We flag it as a direction, not a derivation.

Second, and closest of all, is the **Montevideo interpretation** of Gambini, Porto, and Pullin [36, 37], which shares the core intuition that *real clocks produce real loss of coherence*. There, the fundamental imprecision of any physical clock means evolution referred to a real clock obeys a master equation with a universal, clock-induced decoherence rate, so off-diagonal terms decay without a collapse postulate. MCI agrees that decoherence is a physical, clock-mediated process rather than a bookkeeping device, but differs in three specific ways. (i) *Source.* Montevideo's loss of coherence is fundamental and universal — a property of time itself referred to imperfect clocks — whereas MCI's is environmental and dissipative: the Adler equilibration of an *opened* system to a bath (§2.3, §3.2), absent for a closed system however imprecise its clock. (ii) *Outcome selection.* Montevideo, like environmental decoherence, yields an improper mixture and does not by itself select a single outcome; MCI adds a beable — the per-run background configuration (§3.3) — to carry the decohered state into one basin. (iii) *Preferred frame.* Montevideo is built to respect general covariance; MCI openly carries a preferred frame entering through Stage 2 (§8). The honest summary is that MCI is a dissipative, single-world cousin of Montevideo: it relocates the clock-decoherence idea from a fundamental temporal imprecision to a concrete environmental synchronization, then pays for definite outcomes with a nonlocal beable and a preferred frame that Montevideo does not require.

Finally, **relational quantum mechanics** [35] shares MCI's rejection of any privileged observer but takes the opposite metaphysical route: RQM is perspectival and denies absolute, observer-independent states, whereas MCI is resolutely single-world and $\psi$-ontic, with one actual field configuration per event. We note the shared anti-anthropocentrism but claim no kinship beyond it.

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## 8. Open problems: what would make this a theory

Two problems separate a single-world *picture* from a single-world *theory*, and the framework's status depends entirely on them:

1. **The Born measure.** A derivation that the long-run basin frequencies are $|\alpha|^2$ — closing both the equal-energy restriction and the "why squared" gap. The missing piece is specifically *typicality*, not basin geometry: the dissipative lock supplies *equivariance* — the weight of the measured projector is conserved, so a $|\cdot|^2$ measure granted at preparation is carried intact to the outcome (Bohm's quantum-equilibrium status) — but the *geometric* basin volume is 50/50, independent of $|\alpha|^2$, so Born cannot come from the attractor geometry and must instead be carried by the measure placed on the selecting ensemble. Equivariance itself, moreover, is not automatic: it requires the Stage-2 coupling to be non-demolition for the *measured* observable, and §3.7 shows the electromagnetic coupling is non-demolition for the charge/position pointer basis but *not* for the chiral interference channel — there it relaxes and erases the prepared weight, so the only coupling that would make the equivariance step hold for the chiral binary is gravitational (§3.7, §7.4). Candidate routes include deriving the relevant symmetry from the U(1) invariance of the chiral phase under global phase redefinition, and the stochastic-mechanics tradition (Nelson [19]); we flag that the latter carries a known obstruction (the Wallstrom single-valuedness condition) and that stochastic electrodynamics has historically reproduced only part of quantum mechanics. A further route is *relaxation to quantum equilibrium*: under a suitably mixing sub-quantum dynamics an arbitrary initial distribution coarse-grains toward $|\psi|^2$, so the Born measure becomes an equilibrium rather than a postulate [42, 43]. This is the natural reading of the framework's serial background-scrambling — each boundary interaction resamples the unbiased background — but it has been shown for standard Bohmian dynamics, not for MCI's dissipative, boundary-localized selection, so whether the same H-theorem holds here, and with what coarse-graining, remains open.
2. **A dynamical law for the selection.** An explicit equation by which the background-field configuration drives the basin selection — Bohm's analog is the guidance equation. The *ontological* locus of the nonlocality is now identified (§7.5): it is the non-separable field configuration of the entangled pair, not the local thermal reference. What remains is the *dynamical* statement — how that configuration updates across a spacelike interval — together with an explicit proof that the update preserves no-signaling, which is not automatic once the law is written independently of the quantum statistics it must reproduce.

Beyond these, the framework's relation to Lorentz invariance must be stated openly, because this revision settles it rather than leaving it ambiguous. MCI carries a **real preferred frame**: the rest frame of the bulk/vacuum reference — in the conservative reading, the local mass–energy rest frame. Whether it is instead a *cosmic* frame (the CMB rest frame is the natural candidate) is an open, testable question: only a cosmic frame produces a sidereal signature, and the LIGO sidereal companion test is designed to discriminate the two. We do not pretend the framework is fully relational. What disciplines the commitment is *where* the frame enters. The Stage-1 chiral dynamics is exactly Lorentz covariant; the preferred frame appears only through Stage-2, the dissipative lock to the bulk — so Lorentz invariance is not broken fundamentally but **spontaneously, by the measurement event**, which selects the bulk frame. Away from measurement, ordinary matter couples to that frame, beyond the electromagnetic locking itself, only **gravitationally** — through the potential differences that produce the redshift channel of §6. Because gravity is weaker than the electromagnetic, strong, and weak interactions that govern particle physics by tens of orders of magnitude, the residual anisotropy this imposes on ordinary, non-gravitational particle interactions is suppressed in proportion — which is why the precise clock-comparison and Michelson–Morley-type bounds have not registered it.

Two caveats keep this honest. First, the *locus* of the vacuum coupling is itself a hypothesis — the **Vacuum Preferred-Frame Hypothesis (VPFH)**: that the quantum vacuum carries a physical rest frame (plausibly the cosmic/CMB frame) and that its coupling to matter is *anisotropic* in that frame. By §2.5 this coupling acts at boundaries, not in free flight, so its signature is a **sidereal modulation of the Stage-2 (boundary) coupling** as the apparatus orientation sweeps past the frame's axis (the CMB apex) — a second-moment effect, a loss of fringe visibility or squeezing, never a coherent phase shift, which the Michelson–Morley bounds already exclude. Whether the coupling additionally reaches into *coherent, in-flight propagation* — a stronger, velocity-dependent, zitterbewegung-scale form that is **not** gravitationally suppressed — is a further and more aggressive claim. The VPFH is not a commitment of this paper but of the companion analyses, which supply its two tests: the LIGO sidereal-anisotropy proposal probes the boundary anisotropy directly, and the fringe-visibility-versus-$\gamma$ proposal [26] probes the stronger in-flight form (and reads null if the coupling is confined to boundaries). A null in both leaves MCI with only the weak gravitational coupling; a positive result promotes the preferred frame to a boundary anisotropy, and in the in-flight case to a laboratory-velocity effect. Second, a *quantitative* demonstration that the surviving gravitational-strength anisotropy lies below the established preferred-frame bounds is still owed: we have given the reason it should be suppressed, not a proof that it clears every constraint. A fully Lorentz-covariant formulation of the Stage-2 selection — the explicit covariant locus of the nonlocality — therefore remains a third open problem, alongside the Born measure and the dynamical selection law.

This also locates the §6.2 gap precisely. The polarization-dependent coupling of measurement to the local reference, on which the one candidate prediction rests, is a postulate we cannot derive from standard QED (Appendix C) and is non-covariant as written — it is this same preferred-frame commitment, now made explicit, showing up at the level of a single measurement. On distinguishability: §6 is the one experimental handle, and even it tests that added postulate rather than the framework's core; the remainder of the framework is, by construction, consistent with standard quantum mechanics rather than distinguishable from it. We state that plainly rather than imply otherwise.

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## Appendix A: The clock-as-hidden-variable model is sub-classical

This result is the formal justification for confining the framework to the measurement mechanism (§1.3). Consider the simplest local model in which a single classical clock phase per particle determines the click probability by Malus's law, P(+1 | n̂, $\varphi$) = $\cos^2$(($\varphi$ − $\theta$_n̂)/2). Evaluated for an EPR pair with a shared initial phase (`tests/bell_phase.py`), this returns

$$E_{\mathrm{Malus}}(a,b) = -\tfrac{1}{2}\cos(a-b), \qquad \mathrm{CHSH}_{\mathrm{Malus}} \leq \sqrt{2} \approx 1.414.$$

This is **sub-classical**: it falls below even the local-hidden-variable bound of 2, and far below the quantum value $2\sqrt{2}$. A phase-clock model that treats the clock as a local variable determining the outcome therefore cannot reproduce — cannot even reach the classical bound for — the observed Bell violations. The full spinor (or polarization) structure of standard quantum mechanics is required to recover −cos(a − b). This is why the framework attributes the correlations to standard quantum mechanics and reserves synchronization for the local measurement mechanism: the alternative is provably ruled out by the framework's own numerics.

## Appendix B: Spin-statistics (consistency check)

MCI does not derive spin-statistics; it inherits it from the underlying Dirac theory and shows where the fermion sign lives. The chiral pair is, as a representation, the (½,0) ⊕ (0,½) Lorentz representation; under $2\pi$ rotation each Weyl half acquires −1, and by the standard continuous-deformation argument exchange inherits the same sign. In the Madelung decomposition $\psi$_X = $\sqrt{\rho_X}$ e^{i$\varphi$_X} $\chi$_X, the amplitude $\rho$_X (real, ≥ 0) and the phase $\varphi$_X (a real ODE variable, symmetric under particle exchange) carry no sign; only the spinor frame $\chi$_X does. `tests/spin_statistics.py` verifies that the exchange amplitude is −1 to numerical precision across six decades in v/c, and that the Kuramoto phase ODE alone is exchange-symmetric and sign-free. This is a **consistency check** — it confirms the framework's chiral-pair commitment is compatible with the standard theorem; it could have exposed an internal inconsistency in the reframing but is not, and could not be, an independent derivation.

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## Appendix C: The redshift cancellation is structural — a Hamiltonian sketch

This appendix supports §6.2: under a standard light–matter Hamiltonian the gravitational redshift enters the polarization correlation as a common-mode phase and cancels, so the candidate effect of §6 requires the additional postulate $H'$ rather than following from the framework plus standard optics.

**Setup.** A polarization-entangled pair in wavepackets of spectral width $\Delta\nu$ about $\omega_0$,

$$|\Psi\rangle = \frac{1}{\sqrt{2}}\int d\omega_A\, d\omega_B\, f(\omega_A)f(\omega_B)\big[|H,\omega_A\rangle|H,\omega_B\rangle + |V,\omega_A\rangle|V,\omega_B\rangle\big].$$

Each detector is a bulk-bound electron with the standard dipole coupling $H^A_{\text{int}} = -\mathbf{d}_A \cdot \mathbf{E}_A(t)$, referenced to its local potential Φ_A.

**Factorization.** The absorption amplitude for analyzer outcome $|a\rangle = \cos a\,|H\rangle + \sin a\,|V\rangle$ separates as

$$M_A(a) = \underbrace{(\varepsilon_a \cdot \mathbf{d}_{eg})}_{\text{polarization projection}} \times \underbrace{\int dt\, e^{i(\omega_0^A-\omega)t}(\cdots)}_{T_A\text{: temporal/energy factor}}, \qquad T_A = |T_A|\, e^{i\phi_A},$$

where the local-clock phase φ_A — carrying the redshift, since ω0^A is referenced to Φ_A — lives entirely in the temporal factor T_A and is *independent of the analyzer angle a.*

**Cancellation.** The joint amplitude is

$$\langle a,b|\Psi\rangle = \frac{1}{\sqrt{2}}\, T_A T_B\, (\cos a \cos b + \sin a \sin b) = \frac{1}{\sqrt{2}}\, T_A T_B \cos(a-b),$$

so the coincidence probability is

$$P(a,b) \;\propto\; |T_A|^2\, |T_B|^2\, \cos^2(a-b).$$

The phases φ_A, φ_B — hence δφ_grav — appear only inside the angle-independent prefactor $|T_A|^2|T_B|^2$ and drop out of the normalized correlation. CHSH = $2\sqrt{2}$, independent of $\Delta\nu$ and $\Delta\Phi$. This is the null of §6.2.

**What would be needed.** A surviving effect requires the HH and VV amplitudes to carry *different* phases, $\phi_{HH} \neq \phi_{VV}$ — a polarization-dependent local-clock phase. At a single detector H and V are absorbed at the same potential, so $\phi_A^H = \phi_A^V$ and the requirement fails term by term; the four apparatus elements that might evade this are tabulated in §6.2, and each either references both branches to the same potential or supplies only a unitary (birefringent) phase that is recalibratable and linewidth-independent. The minimal term that would produce the effect, $H' \sim g\,(\mathbf{d}\cdot\hat{\mathbf{H}})\,\phi_{\text{bulk}}$, is absent from QED, breaks the rotational covariance of the dipole interaction, and is tied to absolute φ_bulk. The §6 measurement is therefore a test of this postulate, not a consequence of standard theory.

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## Author Contributions and AI Disclosure

This paper was developed through an extended collaboration between the listed human author (J. Bramble) and Claude (Anthropic) across multiple model versions during 2026. The human author contributed the originating physical intuitions — that interacting quantum systems tend to synchronize, that measurement is re-synchronization to a detector bulk, that NMR/MRI T1/T2 relaxation is recovery toward a bulk-defined equilibrium rather than randomization, and that detectors at different gravitational potentials might differ in their measurement dynamics — together with physical judgment, redirection, and validation against intuition at each step. Claude contributed the explicit derivations (the Madelung reduction of §2.2, the normal-mode analysis, the Bell-decomposition numerics), the Python verification code, the systematic comparison with existing interpretations, and drafts of the prose. The analysis throughout separates the unitary closed-system regime from the dissipative open-system one, identifies the Zitterbewegung beat with the positive/negative-energy normal-mode splitting, avoids unnormalized collective-rate estimates, establishes by an explicit Hamiltonian argument (Appendix C) that the linewidth-dependent gravitational effect does *not* follow from the framework plus standard QED but requires an added, non-covariant postulate — demoting it from a derived prediction to a test of that postulate — and places the interpretation honestly within the single-world, nonlocal family.

Per current editorial guidelines (Nature Portfolio, Springer Nature, and others), large language models do not satisfy authorship criteria, because authorship carries accountability that cannot be applied to an LLM. We accept this framing — structurally analogous to the attending physician's ultimate accountability regardless of consultative input — and J. Bramble assumes full responsibility for the correctness, originality, and integrity of all content, including any errors in the derivations or interpretations.

Simulation code and numerical verification: https://github.com/rayolddog/DiracKuramotoFramework

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